

Recitaion 02

Input, processing and output

Exercises 1 – Numeric values and operations

Evaluate the following expressions and specify their type.

5 * -6		34 / 10	
(34 // 10) * (1.0 + 2)		34 % 10	
23 % 4.0		25 + 30 / 6	
-7 % 3		100 - 25 * 3 % 4	

Exercise 2 - Looking for Prints

Write out the output of each line in the space provided. If the line results in an error, write "error".

<code>print(5 + "5")</code>	<code>print(5 + int("5"))</code>
<code>print(str(5) + int("5"))</code>	<code>print(15 % 2)</code>
<code>print("hey" + "yo")</code>	<code>print(2 * "go")</code>
<code>print("yo" * 4.0)</code>	<code>print("hi" + there)</code>
<code>print("go" * 2 + "!" * 2)</code>	

Exercises 3 - Escape sequence

Write the output, or Error if an error occurs.

<code>print("Welcome to \'ICP\'")</code>	
<code>print("Welcome to \"ICP\"")</code>	
<code>print('Welcome to "ICP"')</code> <code>print('Welcome to \"ICP\"')</code>	
<code>print("Welcome to \"ICP\"")</code>	
<code>print("Today is\t\"Tuesday\"\nWelcome to\t\"ICP\"")</code>	

Exercise 4 | Variable assignment

Consider the following variable assignments:

Left = "Lady"

Middle = "Jack"

Right = "Ace"

Figure out where the 'Lady' is (Left, Middle, Right) after running the following programs:

Left, Right, Middle = Left, Middle, Right
Left, Right, Middle = Middle, Right, Left
Left, Middle, Right = Middle, Right, Left
Left, Middle, Right = Middle, Right, Left Left, Middle, Right = Middle, Right, Left
Left, Middle, Right = Middle, Right, Left Left, Middle, Right = Middle, Right, Left Left, Middle, Right = Middle, Right, Left
Left, Middle, Right = Right, Right, Right Left, Middle, Right = Left, Left, Left Left, Middle, Right = Middle, Middle, Middle

Exercise 5 | Print arguments

<pre>a = 5 print("a", a, sep=' = ')</pre>	
<pre>a = 5 print("a = ", a, sep="0000")</pre>	
<pre>a = 5 print("a = ", a, sep="", end="")</pre>	
<pre>a, b = 'b', 'a' print(a, 'b', sep='>>>', end='\t') print('a', b, sep='***', end='\n\n') print(a, b, sep='\t')</pre>	

Exercise 6 | String formatting

<pre>print(f"{12:8d}")</pre>	
<pre>print(f"{'12':0>8}")</pre>	
<pre>num = 12.34567890 print(f"{num:8.3f}")</pre>	
<pre>print(f"{12.34567890:<08.3f}")</pre>	
<pre>name = 'dolphin' res1 = f'{name:->10}' res2 = f'{name:<10.3}' print(res1) print(res2)</pre>	

Exercise 7 | Sales Prediction

A company has determined that its annual profit is typically 23 percent of total sales. Write a program that asks the user to enter the projected amount of total sales, and then displays the profit that will be made from that amount.

Exercise 8 | Celsius to Fahrenheit Temperature Converter

Write a program that converts Celsius temperatures to Fahrenheit temperatures. The formula is as follows:

$$F = \frac{9}{5}C + 32$$

The program should ask the user to enter a temperature in Celsius, then display the temperature converted to Fahrenheit.

Exercise 9 | Ingredient Adjuster

A cookie recipe calls for the following ingredients:

- 1) 1.5 cups of sugar
- 2) 1 cup of butter
- 3) 2.75 cups of flour

The recipe produces 48 cookies with this amount of the ingredients.

Write a program that asks the user how many cookies he or she wants to make, and then displays the number of cups of each ingredient needed for the specified number of cookies.